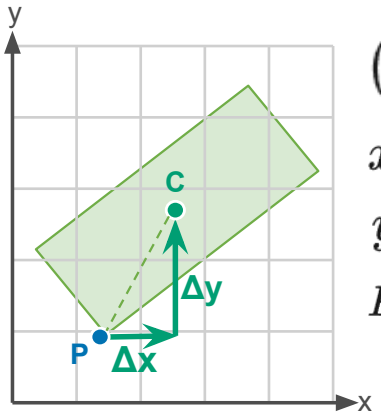
## (a) Offset head



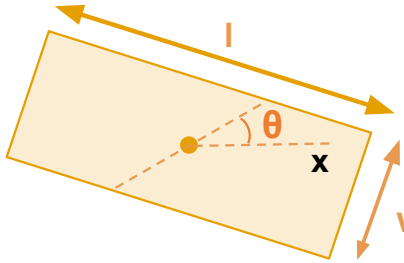

$$(\Delta x, \Delta y)$$

$$x_c = x_p + \Delta x$$

$$y_c = y_p + \Delta y$$

$$P = (x_{\min} + is_x, y_{\min} + js_y)$$

## (b) Size head

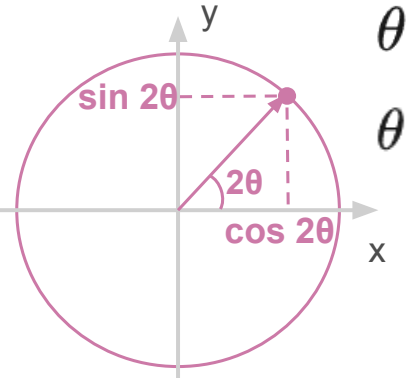



$$(\log w, \log l)$$

$$t_w = \log w, \quad t_l = \log l$$

$$w = \exp(t_w), \quad l = \exp(t_l)$$

## (c) Yaw head

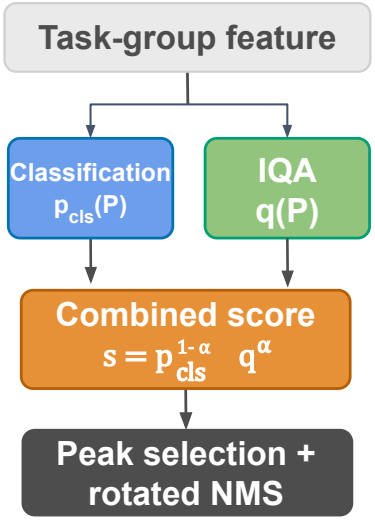



$$(c_2, s_2) = (\cos 2\theta, \sin 2\theta)$$

$$\theta \equiv \theta + \pi s$$

$$\theta = \frac{1}{2}\operatorname{atan2}(s_2, c_2)$$

## (d) IQA head




$$q(P) = \sigma(a_q(P))$$

$$s(P) = \sqrt{p_{\text{cls}}(P)q(P)}$$

$$q(P) \in (0,1), \quad \alpha = 0.5$$